

Design and Comparative Simulation of First-Order and Second-Order Active Filters for Analog Signal Conditioning Using LTspice

Project Report

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Abstract

Signal conditioning is a fundamental requirement in analog electronic systems to selectively process useful frequency components while suppressing unwanted signals and noise. This project presents the design, simulation, and comparative analysis of first-order and second-order active low-pass and high-pass filters using LTspice. First-order filters were implemented using RC networks followed by voltage-follower operational amplifiers, while second-order filters were realized using the Sallen-Key topology. The circuits were designed for a nominal cutoff frequency of approximately 1 kHz using equal resistor and capacitor values. AC sweep and transient analyses were performed to evaluate the frequency response, attenuation characteristics, phase variation, and time-domain performance of each filter. The influence of resistor and capacitor variations on the cutoff frequency was also investigated to validate the theoretical relationship between the RC time constant and filter response. The simulated results closely matched theoretical expectations, demonstrating the effectiveness of active filter topologies in analog signal conditioning and highlighting the advantages of higher-order filters in achieving improved frequency selectivity.

Problem Statement

Analog electronic systems frequently require signal conditioning to selectively retain useful frequency components while attenuating unwanted signals and noise. Designing such circuits requires an understanding of active filter topologies, operational amplifier configurations, and the relationship between component values and frequency response characteristics. This project aims to design and analyze first-order and second-order active filter circuits using LTspice, enabling a

systematic study of frequency-selective analog signal conditioning through simulation-based evaluation.

Objectives

- Design first-order and second-order active low-pass and high-pass filters using operational amplifiers.
- Investigate the frequency-selective behaviour of active filter topologies through simulation.
- Validate theoretical cutoff frequencies using AC sweep analysis.
- Evaluate transient response for representative input signals.
- Study the influence of resistor and capacitor values on cutoff frequency.
- Compare the performance of first-order and second-order filters with respect to attenuation, phase response and frequency selectivity.
- Develop practical understanding of simulation-driven analog filter design using LTspice.

Theory

Active filters employ operational amplifiers together with resistors and capacitors to achieve frequency-selective signal conditioning while providing high input impedance and low output impedance. Unlike passive filters, active implementations minimize loading effects and maintain predictable filter characteristics, making them suitable for analog signal processing applications.

A **low-pass filter (LPF)** allows frequency components below the cutoff frequency to pass with minimal attenuation while reducing higher-frequency components. Conversely, a **high-pass filter (HPF)** attenuates frequencies below the cutoff while allowing higher-frequency signals to pass.

The cutoff frequency for both first-order and equal-component second-order filters is determined by the RC time constant and is given by

$$f_c = \frac{1}{2\pi RC}$$

For the component values used in this work,

- **Resistance (R) = 15.9 kΩ**
- **Capacitance (C) = 10 nF**

the theoretical cutoff frequency is approximately

$$f_c \approx 1 \text{ kHz}$$

The order of a filter determines the sharpness of its frequency response. A first-order filter exhibits an attenuation rate of approximately **20 dB/decade**, whereas a second-order filter provides a steeper attenuation of approximately **40 dB/decade**, resulting in improved rejection of unwanted frequency components.

To evaluate circuit performance, two simulation techniques were employed:

- **AC Sweep Analysis** to study magnitude response, cutoff frequency, attenuation and phase variation.
- **Transient Analysis** to examine the time-domain response of the filters under sinusoidal excitation.

The simulated responses were compared with theoretical predictions to verify the effectiveness of the designed active filter circuits.

Design Calculations

The cutoff frequency for the designed active filters is given by

$$f_c = \frac{1}{2\pi RC}$$

Choosing

$$C = 10 \text{ nF}$$

And

$$f_c = 1 \text{ kHz}$$

gives

$$\begin{aligned}
 R &= \frac{1}{2\pi f_c C} \\
 &= \frac{1}{2\pi(1000)(10 \times 10^{-9})} \\
 &= 15.9 \text{ k}\Omega
 \end{aligned}$$

Therefore,

- $R_1 = R_2 = 15.9 \text{ k}\Omega$
- $C_1 = C_2 = 10 \text{ nF}$

were selected for all filter implementations.

A capacitor value of 10 nF was selected because it is a standard value commonly used in analog filter design, resulting in a practical resistor value of approximately 15.9 k Ω .

Design Methodology

The active filter circuits were designed to investigate the frequency-selective behaviour of first-order and second-order analog signal conditioning networks using LTspice. To enable a meaningful comparison, all filter configurations were designed for a common theoretical cutoff frequency of approximately **1 kHz** by selecting equal resistor and capacitor values according to the standard cutoff frequency relation.

First-order low-pass and high-pass filters were implemented using passive RC networks followed by a **unity-gain voltage follower**. The voltage follower isolates the filter network from loading effects by providing high input impedance and low output impedance, thereby preserving the designed frequency response.

Second-order filters were implemented using the **unity-gain Sallen-Key topology**, which integrates the operational amplifier within the filter network to achieve a second-order frequency response while maintaining a simple circuit configuration. This topology was selected due to its widespread use in analog signal conditioning applications and its ability to provide improved frequency selectivity with minimal circuit complexity.

The theoretical cutoff frequency for all filter configurations was determined

where

- Resistance, **R = 15.9 kΩ**
- Capacitance, **C = 10 nF**

resulting in $f_c \approx 1 \text{ kHz}$

To investigate the effect of component selection on filter characteristics, additional simulations were performed by proportionally varying the resistor and capacitor values while maintaining equal component ratios. The resulting shift in cutoff frequency was compared with theoretical predictions to validate the dependence of filter response on the RC time constant.

During circuit implementation, multiple operational amplifier models available in LTspice were evaluated. Initial simulations using practical operational amplifier models produced inconsistent responses due to improper power supply configuration. The simulations were subsequently standardized using the **UniversalOpamp2** model powered from **±15 V** supplies, ensuring ideal linear operation throughout the frequency range of interest. This implementation highlighted the importance of appropriate operational amplifier selection and supply configuration in simulation-driven analog circuit design.

Design Parameters:

Parameter	Value
Filter Types	First-order LPF, First-order HPF, Second-order LPF (Sallen-Key), Second-order HPF (Sallen-Key)
Operational Amplifier	UniversalOpamp2
Supply Voltage	±15 V
R1, R2	15.9 kΩ
C1, C2	10 nF
Theoretical Cutoff Frequency	≈1 kHz

Simulation Setup

All circuits were simulated using **LTspice** under identical operating conditions to ensure a fair comparison between the different filter topologies. The

frequency-domain response of each filter was evaluated using AC sweep analysis, while transient analysis was performed to examine the corresponding time-domain behaviour.

For AC analysis, an input source with an AC amplitude of **1 V** was applied, allowing the transfer characteristics to be observed directly. The frequency sweep was performed over a sufficiently wide range to capture the complete passband, transition region and stopband of each filter.

Transient simulations were carried out using sinusoidal input signals at representative frequencies below, near and above the cutoff frequency to illustrate the filtering action under time-domain excitation.

To validate the theoretical design equations, additional simulations were conducted after modifying the resistor and capacitor values. The resulting cutoff frequencies were compared with the expected theoretical values obtained from the RC time constant.

AC Sweep Command: .ac dec 100 10 100k

Transient Command: .tran 0 0.01 0 0.1m

Simulation Parameters:

Simulation	Configuration
Simulator	LTspice XVII
Input Signal (AC Sweep)	AC Amplitude = 1 V
AC Sweep Type	Decade Sweep
Points per Decade	100
Frequency Range	10 Hz – 100 kHz
Transient Input	1 V Sine Wave
Operational Amplifier	UniversalOpamp2
Supply Voltage	± 15 V

Results and Analysis

First-Order Active Low-Pass Filter

Circuit Schematic

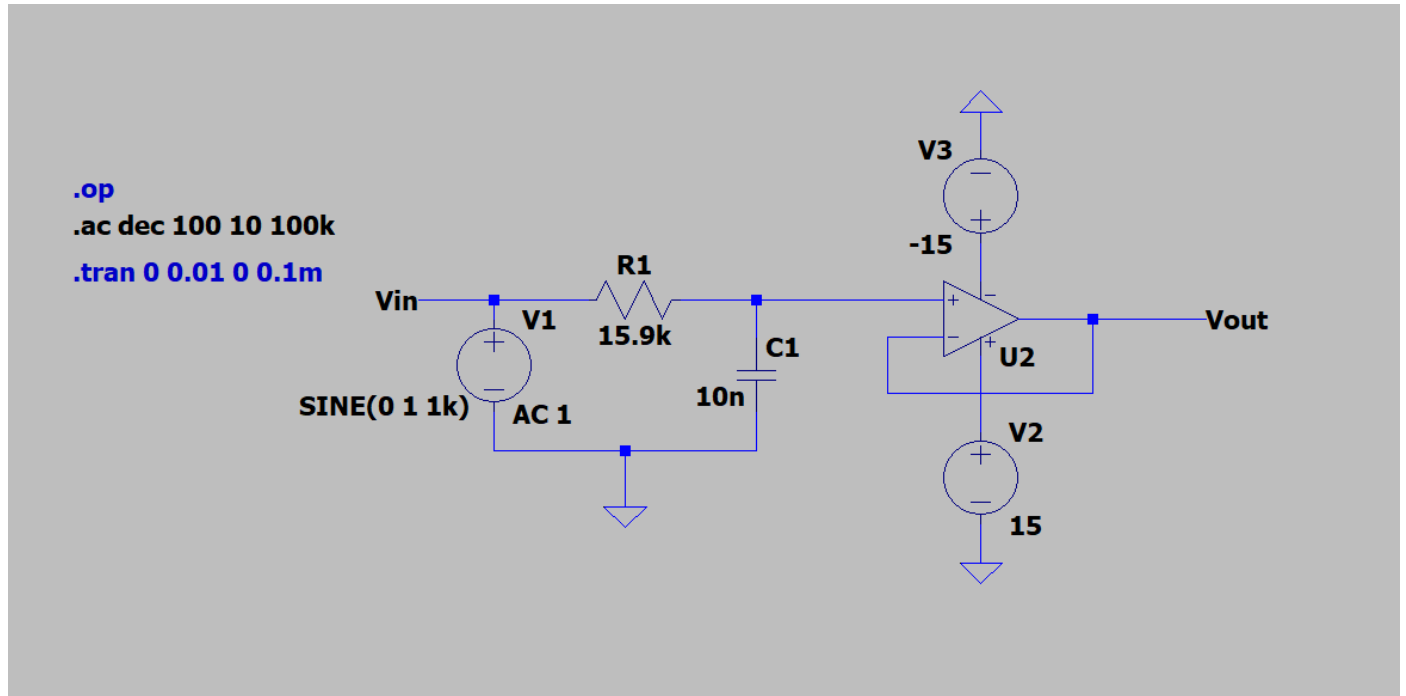


Figure 1. First-Order Active Low-Pass Filter Circuit Implemented in LTspice

AC Sweep Response

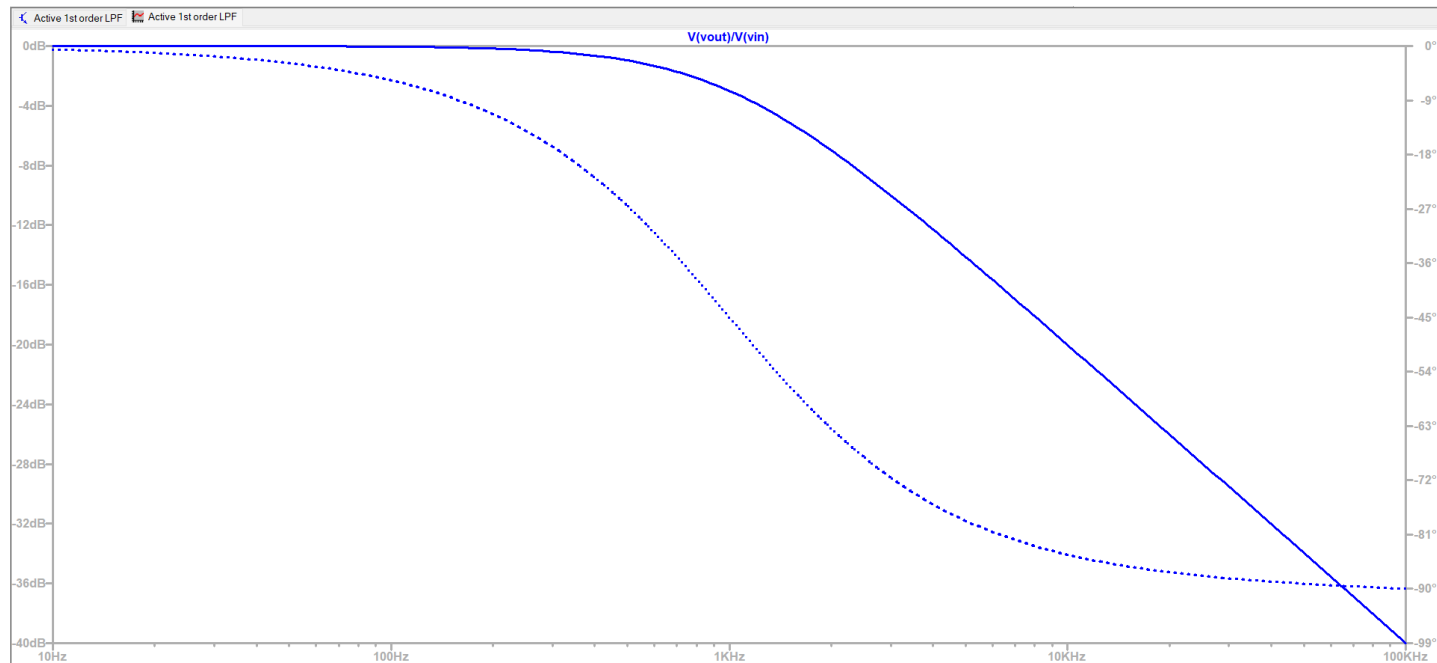
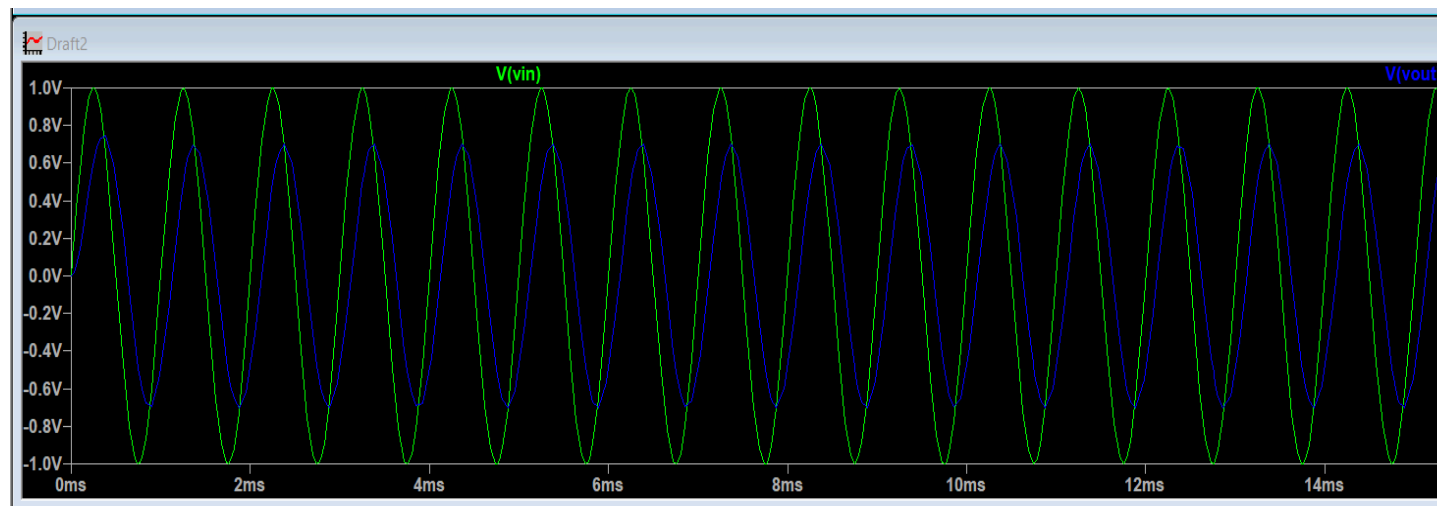


Figure 2. AC Sweep Response of the First-Order Active Low-Pass Filter

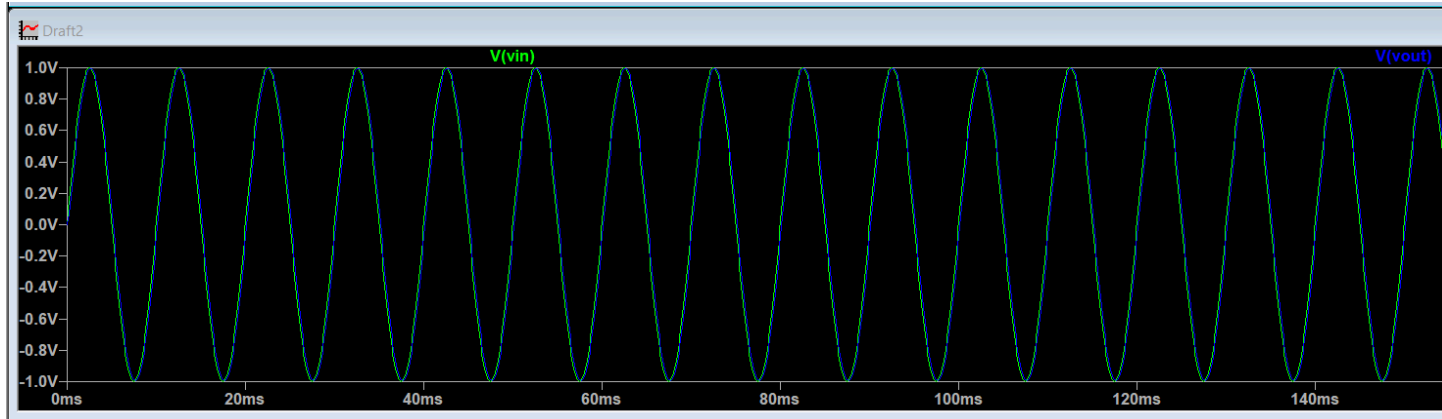
Transient Response

Figure 3. Transient Response of the First-Order Active Low-Pass Filter

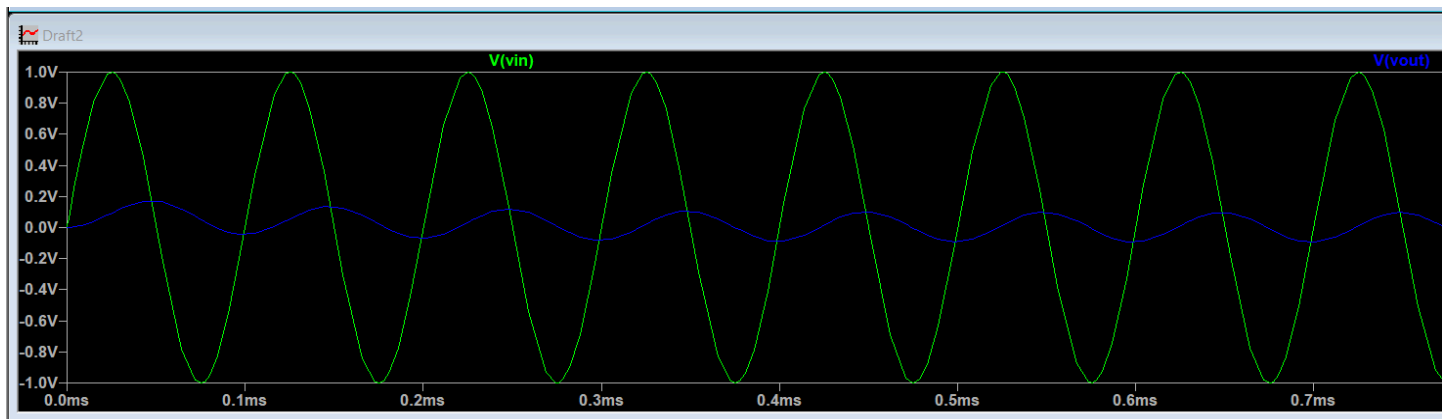
(a) 1kHz



(b) 100Hz



(c) 10kHz



Key Observation

The simulated AC response closely follows the theoretical behaviour of a first-order active low-pass filter. A nearly constant passband gain is maintained up to the designed cutoff frequency of approximately **1 kHz**, beyond which the output magnitude decreases with the expected first-order attenuation characteristic. The phase response exhibits a gradual transition towards **-90°**, confirming the presence of a single dominant pole.

The transient response further demonstrates that low-frequency input signals are reproduced with minimal attenuation, whereas higher-frequency components experience a noticeable reduction in output amplitude along with an increased phase delay.

Design Insight

The voltage follower effectively isolates the RC network from loading effects, ensuring that the designed cutoff frequency closely matches its theoretical value. This configuration demonstrates how active buffering improves the predictability of analog signal-conditioning circuits without altering the intended filter characteristics.

First-Order Active High-Pass Filter

Circuit Schematic

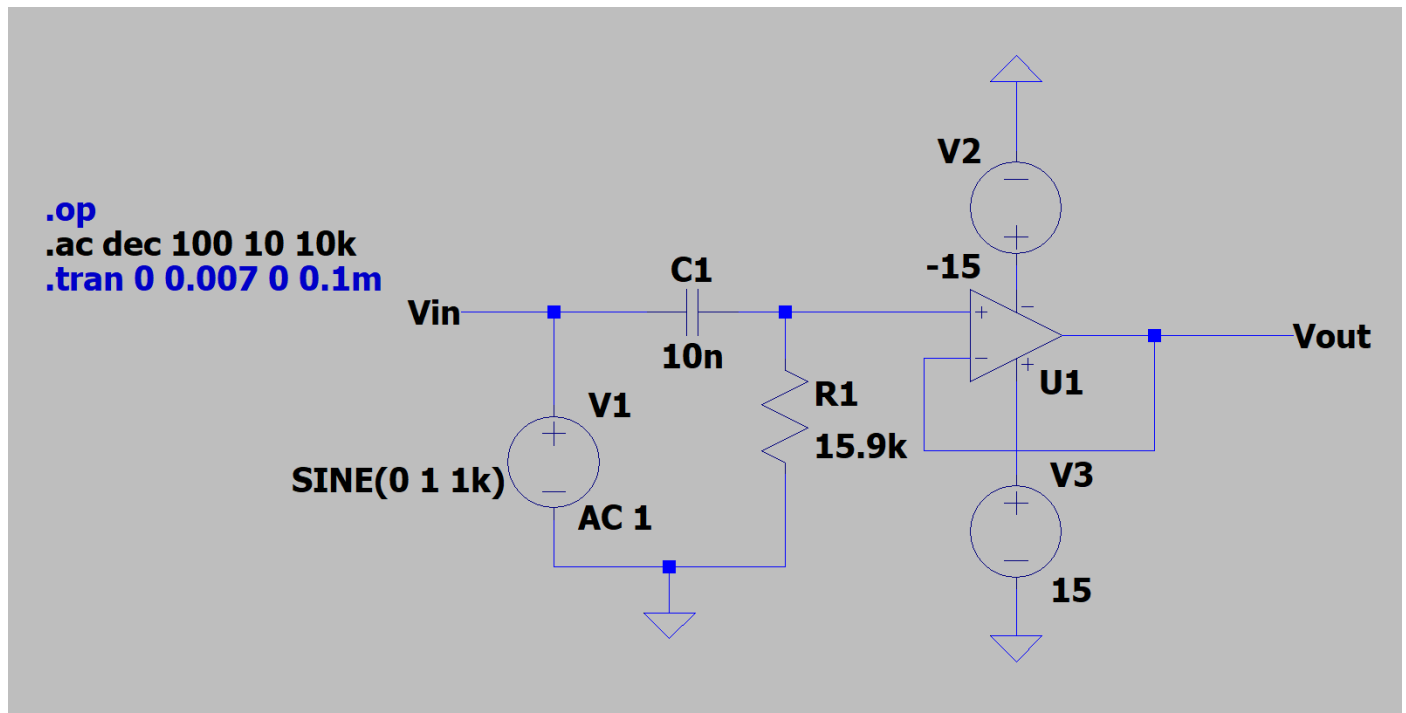


Figure 4. First-Order Active High-Pass Filter Circuit Implemented in LTspice

AC Sweep Response

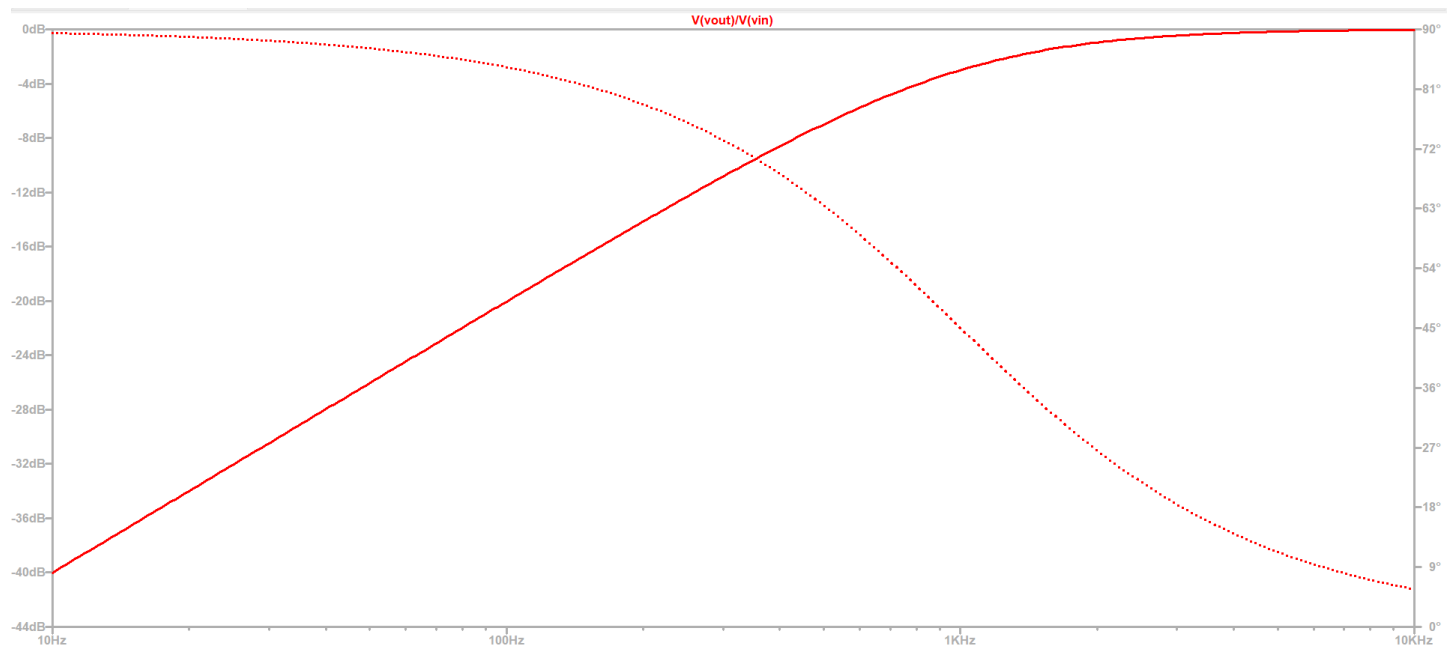
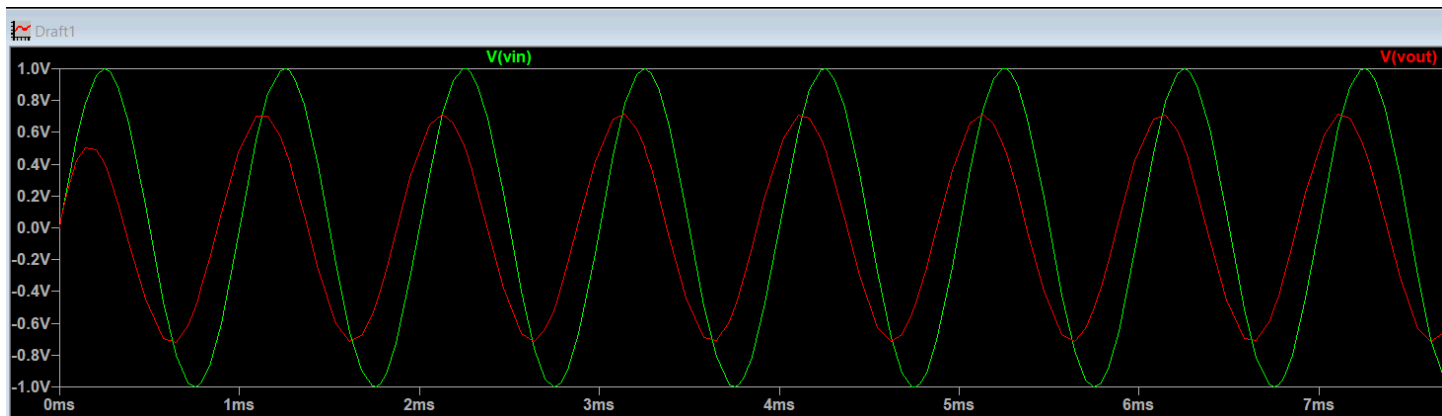


Figure 5. AC Sweep Response of the First-Order Active High-Pass Filter

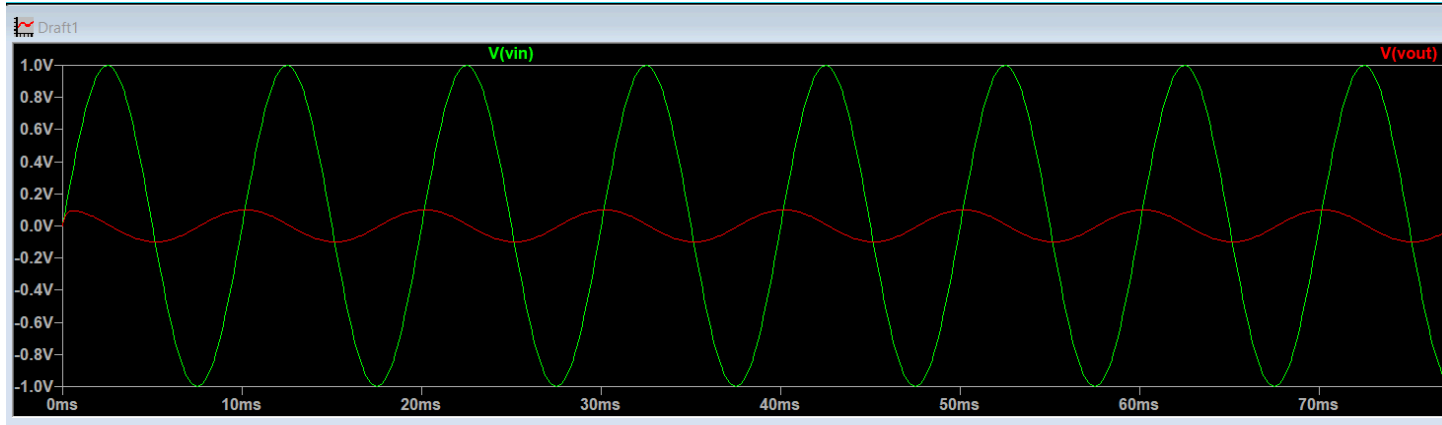
Transient Response

Figure 6. Transient Response of the First-Order Active High-Pass Filter

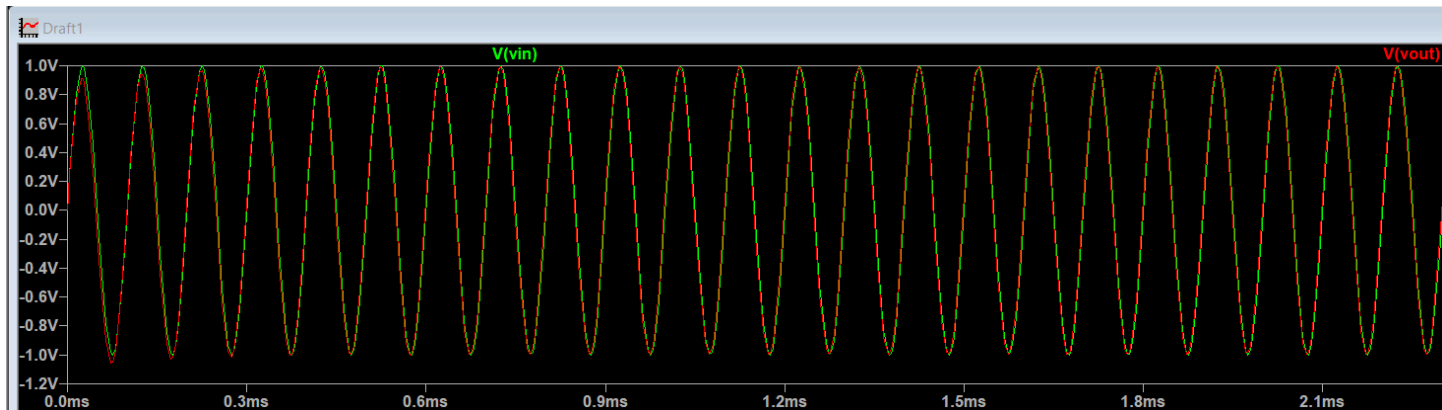
(a) 1kHz



(b) 100Hz



(c) 10kHz



Key Observation

The frequency response confirms the expected behaviour of a first-order active high-pass filter. Low-frequency components are significantly attenuated below the cutoff frequency, while signals above approximately **1 kHz** are transmitted with nearly constant gain. The phase response gradually approaches **+90°** (or equivalently **-270°**, depending on LTspice phase representation), indicating the characteristic phase shift associated with a first-order high-pass network.

The transient response illustrates the suppression of low-frequency signal components while preserving the waveform amplitude at higher input frequencies.

Design Insight

The circuit demonstrates the complementary nature of high-pass filtering in analog signal conditioning. While the low-pass filter suppresses high-frequency noise, the high-pass configuration effectively removes low-frequency disturbances and DC components, making it suitable for signal coupling and baseline correction applications.

Second-Order Active Low-Pass Filter (Sallen-Key)

Circuit Schematic

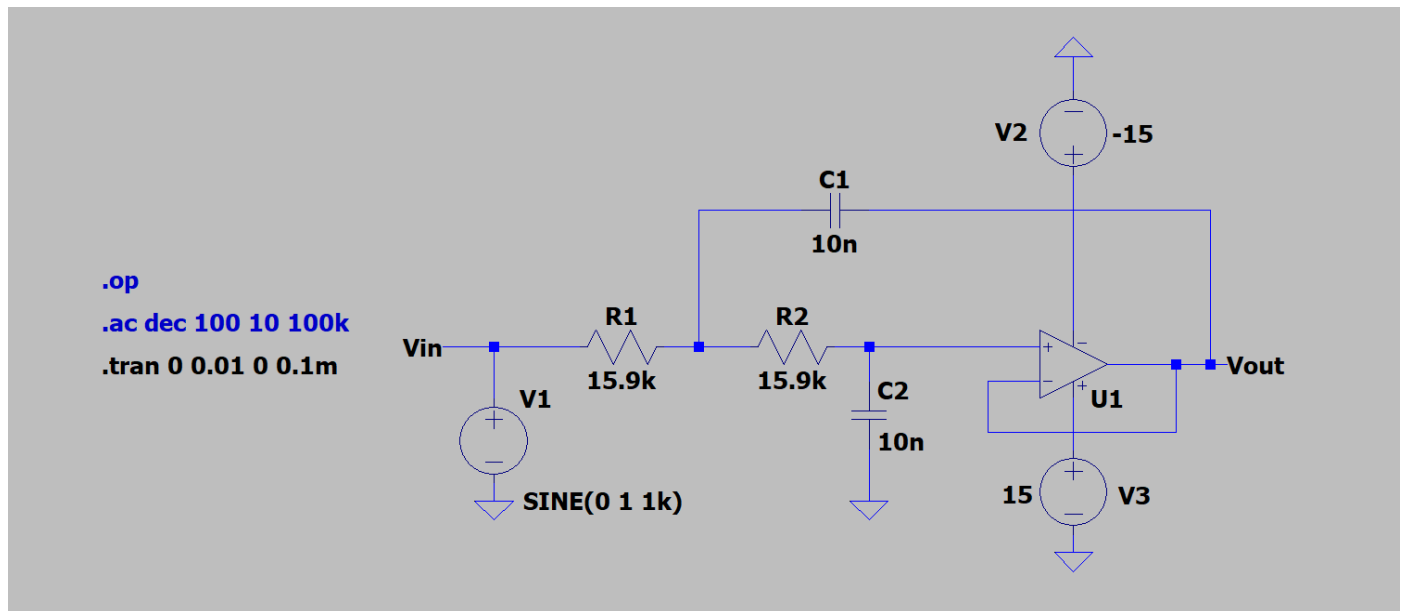


Figure 7. Second-Order Sallen-Key Active Low-Pass Filter Circuit

AC Sweep Response

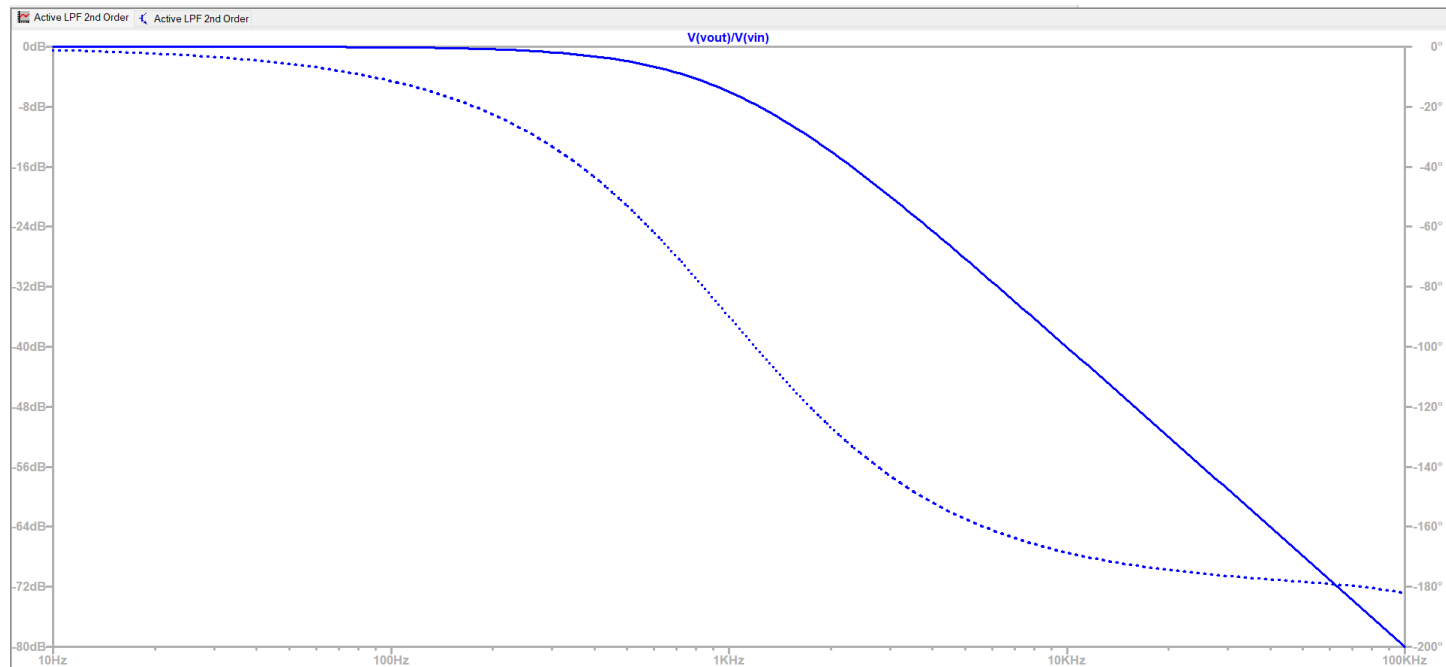
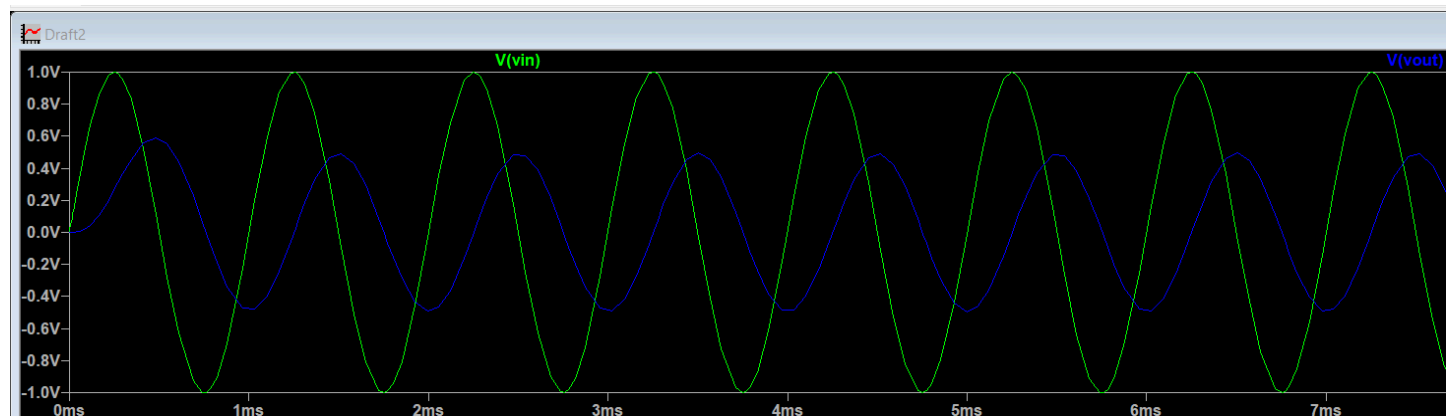


Figure 8. AC Sweep Response of the Second-Order Active Low-Pass Filter

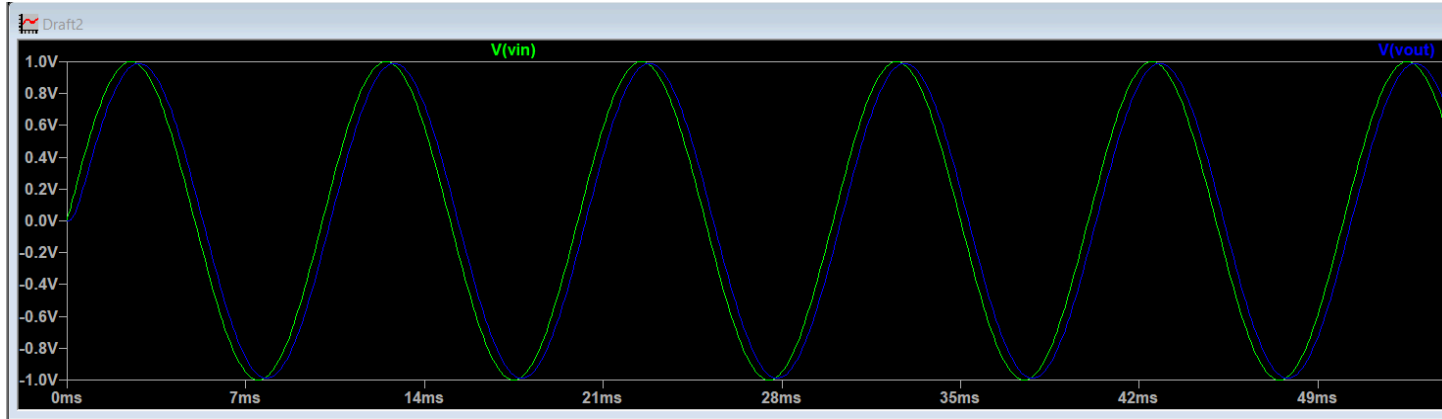
Transient Response

Figure 9. Transient Response of the Second-Order Active Low-Pass Filter

(a) 1kHz



(b) 100Hz



(c) 10kHz

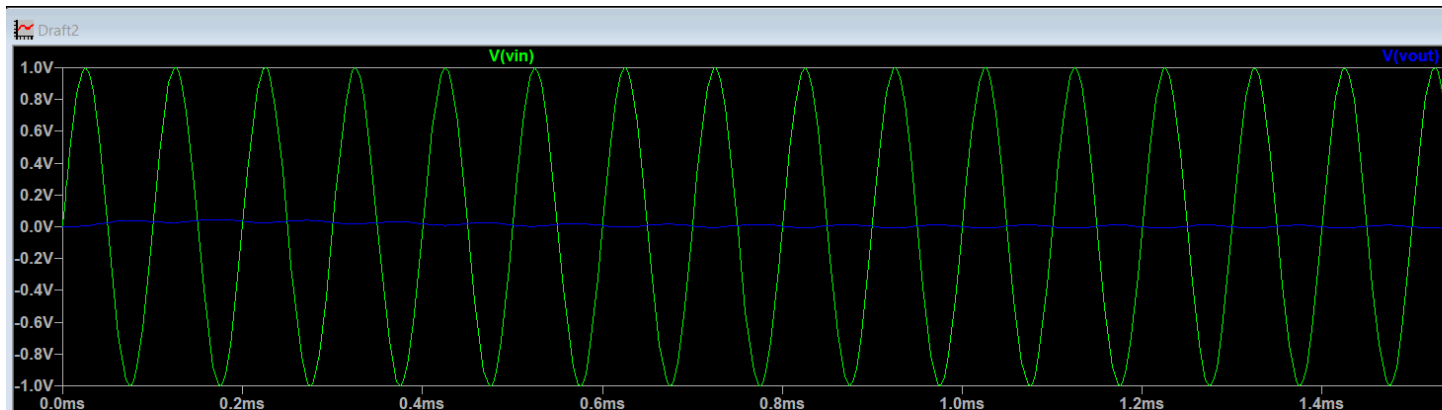


Figure 9.c. Transient Response of the Second-Order Active Low-Pass Filter for a sine wave of frequency 10kHz

Key Observation

The second-order Sallen-Key low-pass filter maintains the designed cutoff frequency while exhibiting a significantly sharper transition between the passband and stopband compared to the first-order implementation. The magnitude response shows an attenuation approaching **-40 dB/decade** beyond the cutoff frequency, accompanied by a phase shift approaching **-180°**, confirming the expected second-order behaviour.

The transient response indicates improved suppression of high-frequency signal components while maintaining stable operation within the passband.

Design Insight

Increasing the filter order enhances frequency selectivity without changing the target cutoff frequency. The Sallen-Key topology achieves this improvement using a single operational amplifier and a simple passive network, illustrating an effective approach for implementing higher-order analog signal-conditioning circuits.

Second-Order Active High-Pass Filter (Sallen-Key)

Circuit Schematic

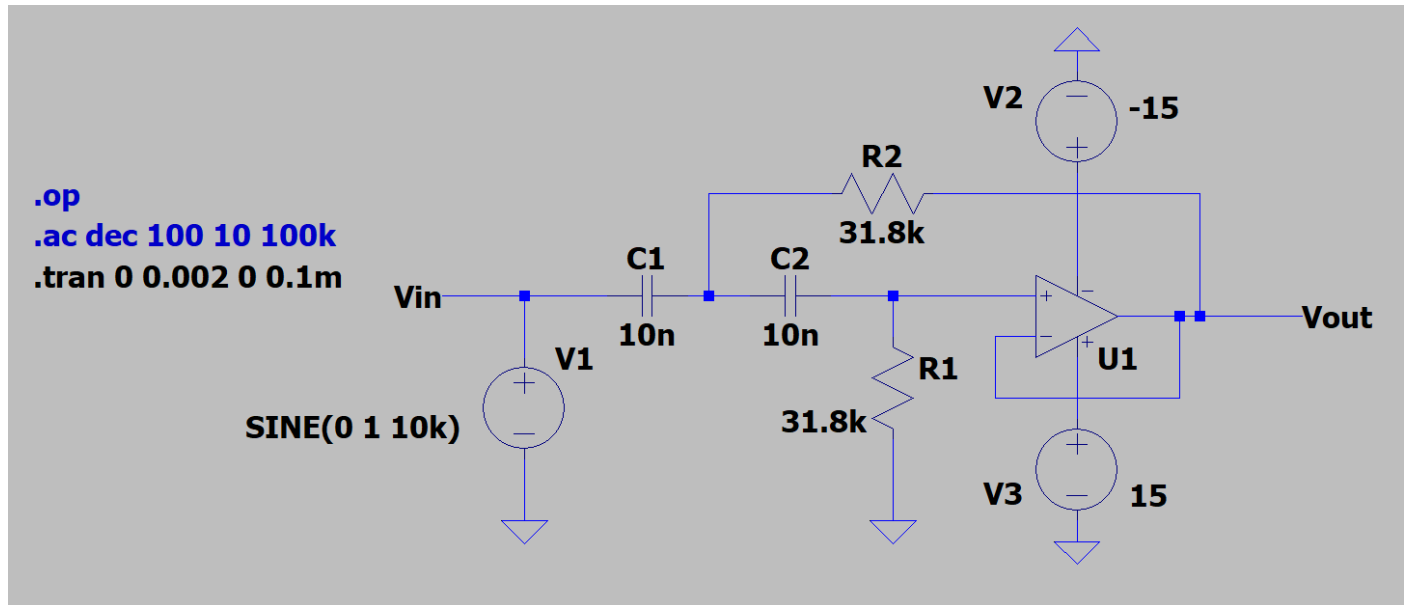


Figure 10. Second-Order Sallen-Key Active High-Pass Filter Circuit

AC Sweep Response

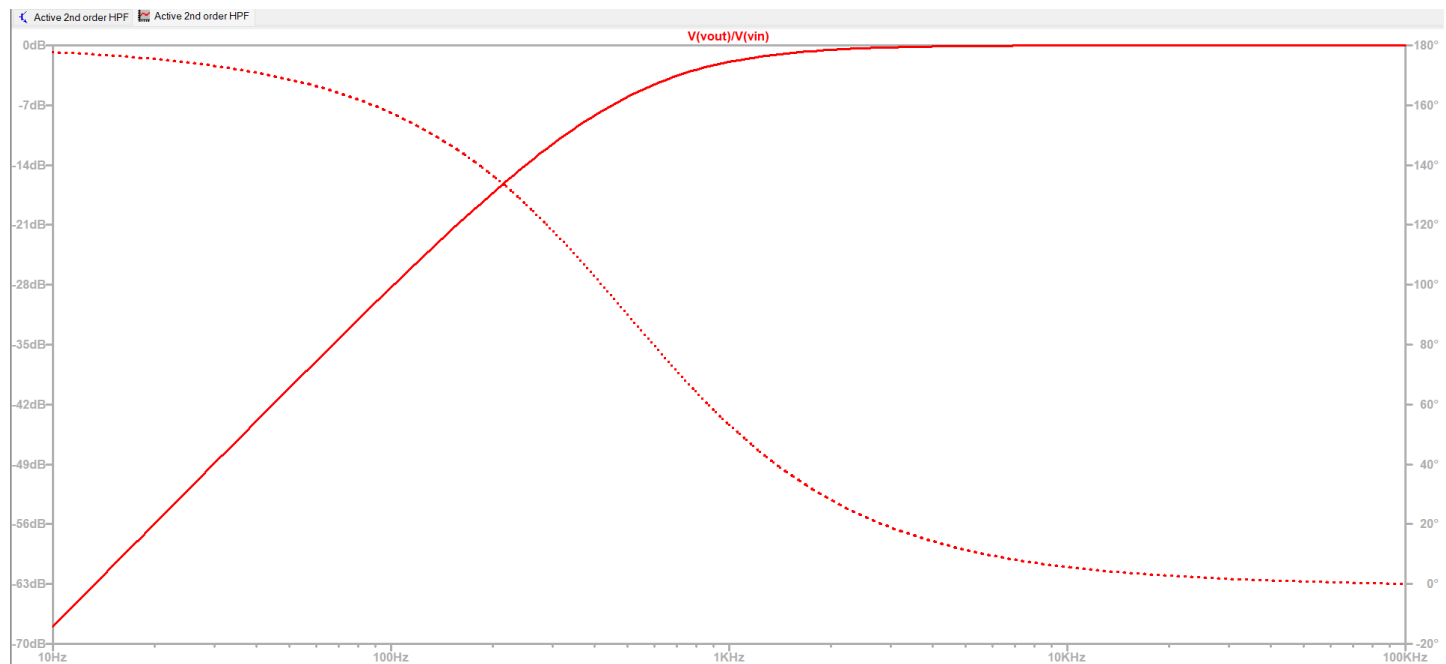
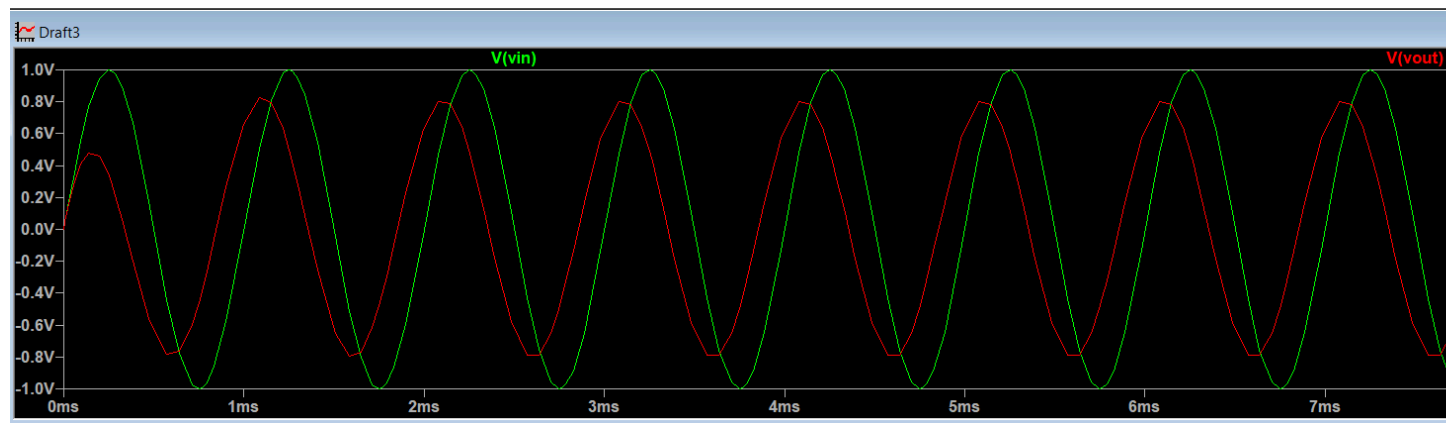


Figure 11. AC Sweep Response of the Second-Order Active High-Pass Filter

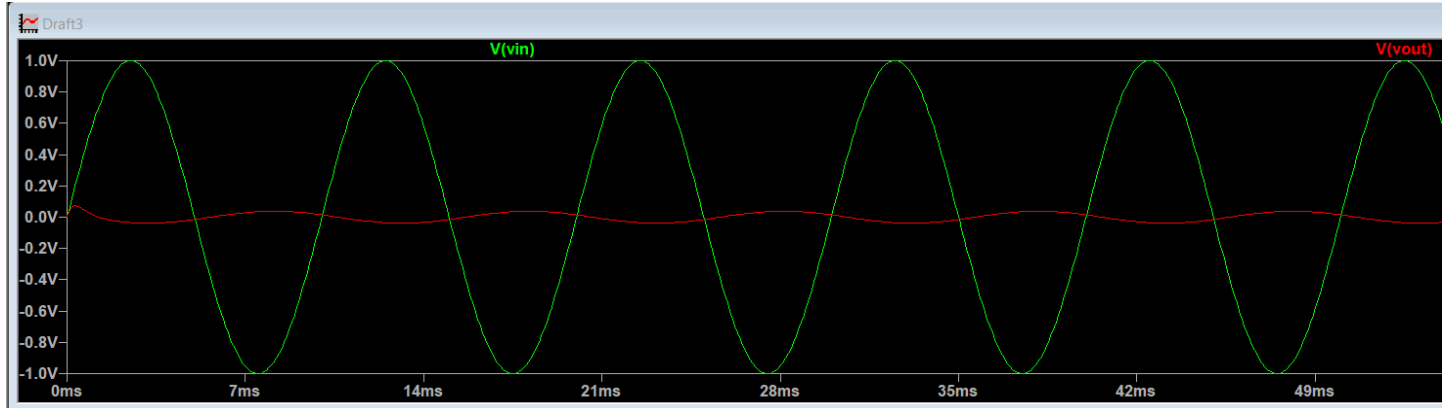
Transient Response

Figure 12. Transient Response of the Second-Order Active High-Pass Filter

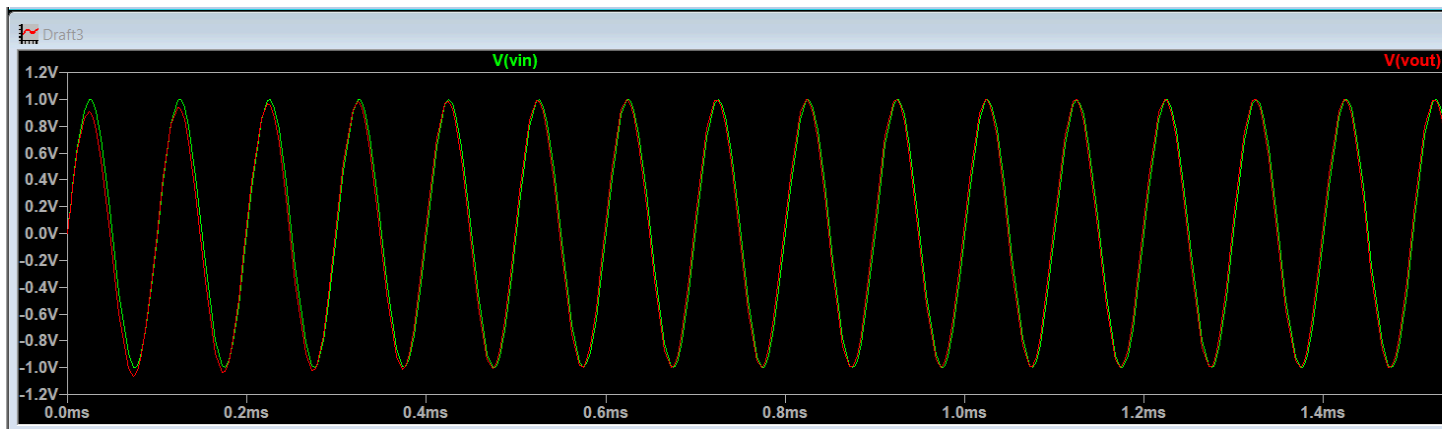
(a) 1kHz



(b) 100Hz



(c) 10kHz



Key Observation

The second-order high-pass filter exhibits strong attenuation of low-frequency components while maintaining a nearly constant passband gain above the cutoff frequency. Compared with the first-order implementation, the transition region is considerably steeper, providing improved rejection of undesired low-frequency signals. The phase response approaches the theoretical second-order limit, demonstrating the increased phase variation associated with higher-order filtering.

The transient analysis verifies that low-frequency input signals are effectively suppressed while higher-frequency waveforms are transmitted with minimal distortion.

Design Insight

The second-order high-pass filter extends the capabilities of the first-order design by providing superior low-frequency rejection while preserving signal integrity within the passband. This characteristic is particularly beneficial in applications requiring effective removal of baseline drift and low-frequency interference before further analog processing.

Comparative Analysis

The designed filters were evaluated to compare the influence of filter order and frequency response characteristics on analog signal conditioning. Although all four circuits were designed for an identical theoretical cutoff frequency of approximately **1 kHz**, their attenuation characteristics, phase response and signal selectivity differed according to the implemented topology.

The first-order filters exhibited a gradual transition between the passband and stopband, making them suitable for applications where moderate frequency discrimination is sufficient. In contrast, the second-order Sallen-Key filters provided a significantly sharper transition region and improved rejection of unwanted frequency components, demonstrating the advantages of higher-order active filter implementations.

This improvement was achieved without changing the desired cutoff frequency, demonstrating that filter order primarily influences the sharpness of frequency discrimination rather than the cutoff location itself.

Similarly, the low-pass and high-pass configurations performed complementary signal-conditioning functions. The low-pass filters preserved lower-frequency components while suppressing high-frequency noise, whereas the high-pass filters attenuated low-frequency components and DC offsets while allowing higher-frequency signals to pass with minimal attenuation.

Comparison of First-Order and Second-Order Filters:

Parameter	First-Order Filter	Second-Order Filter
Filter Order	1	2
Cutoff Frequency	≈ 1 kHz	≈ 1 kHz
Passband Gain	Approximately Constant	Approximately Constant
Roll-off Rate	-20 dB/decade	-40 dB/decade

Maximum Phase Shift	$\pm 90^\circ$	$\pm 180^\circ$
Frequency Selectivity	Moderate	High
Stopband Attenuation	Lower	Higher
Circuit Complexity	Simple	Moderate

Comparison of Low-Pass and High-Pass Filters:

Parameter	Low-Pass Filter	High-Pass Filter
Passband	Low Frequencies	High Frequencies
Stopband	High Frequencies	Low Frequencies
Typical Application	Noise Suppression	DC Blocking and AC Coupling
Signal Conditioning Objective	Removal of High-Frequency Noise	Removal of Low-Frequency Drift
Output Behaviour Above Cutoff	Attenuated	Preserved
Output Behaviour Below Cutoff	Preserved	Attenuated

Discussion

The comparative study demonstrates that increasing the filter order primarily improves frequency selectivity without altering the designed cutoff frequency. The second-order Sallen-Key filters produced a substantially steeper attenuation beyond the cutoff frequency, enabling more effective suppression of unwanted spectral components than the corresponding first-order filters.

The complementary behaviour observed between low-pass and high-pass configurations highlights the flexibility of active filters in analog signal conditioning. Depending on the frequency range of interest, the appropriate topology can be selected to suppress either high-frequency noise or low-frequency interference while preserving the desired signal content.

Overall, the simulation results validate the theoretical relationship between filter order and frequency response characteristics while illustrating the practical advantages of active filter implementations for analog signal-conditioning applications.

Component Variation Analysis

To investigate the influence of component selection on filter characteristics, the resistor and capacitor values were proportionally varied while maintaining equal component ratios within each filter topology. This approach ensured that the filter response characteristics remained unchanged while only the cutoff frequency was shifted according to the RC time constant.

The observed cutoff frequencies closely followed the theoretical relationship confirming that the cutoff frequency is inversely proportional to both resistance and capacitance.

Component Variation and Observed Filter Behaviour:

Component Variation	Expected Cutoff Frequency	Measured Cutoff Frequency	Observed Behaviour
R = 15.9 kΩ, C = 10 nF	1.001 kHz	938.40 Hz	Baseline Response
R = 31.8 kΩ	500.5 Hz	433.17 Hz	Cutoff shifted to lower frequency
R = 7.95 kΩ	2.002 kHz	1.92 kHz	Cutoff shifted to higher frequency
C = 20 nF	500.5 Hz	424.45 Hz	Similar response with lower cutoff
C = 5 nF	2.002 kHz	2.05 kHz	Similar response with higher cutoff

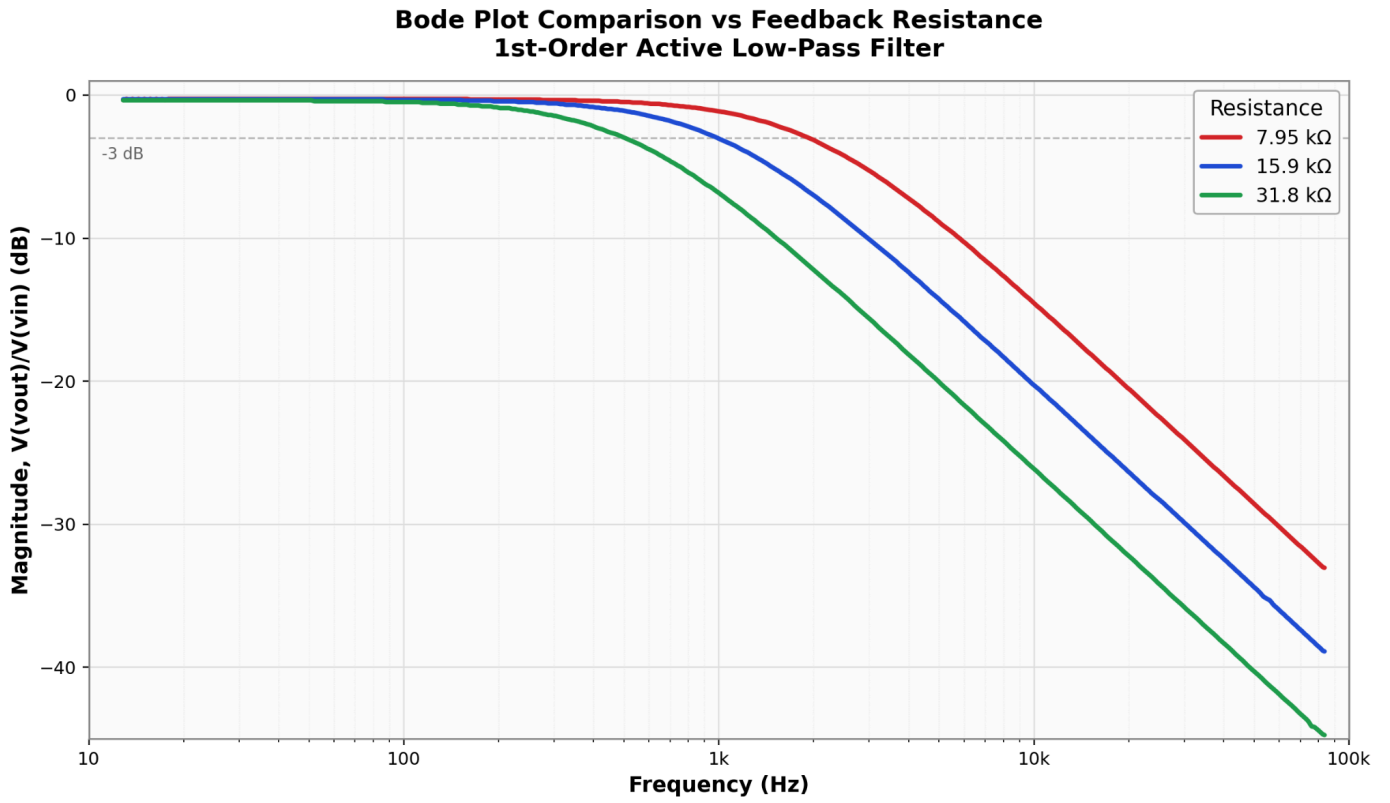


Figure 13. Comparison of AC Responses for Different Resistor Values

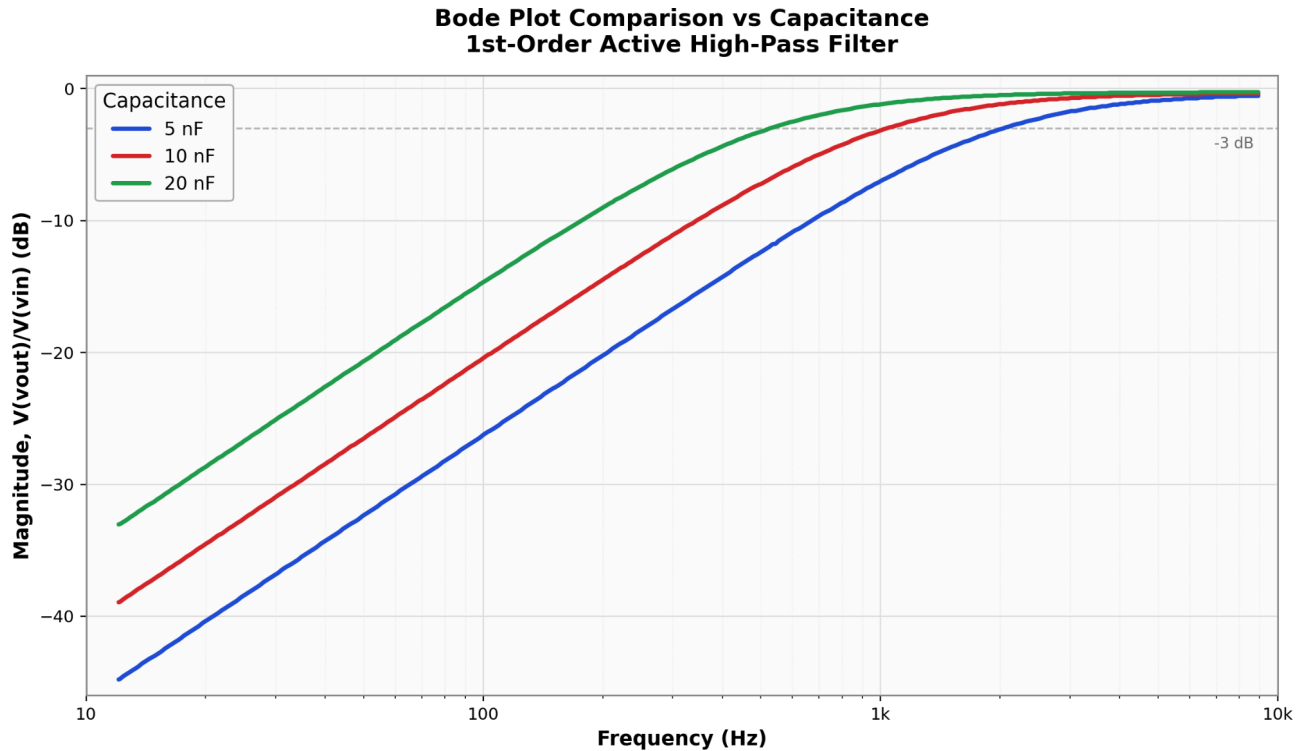


Figure 14. Comparison of AC Responses for Different Capacitor Values

Key Observation

Increasing either the resistance or capacitance increased the RC time constant, resulting in a proportional reduction in cutoff frequency. Conversely, decreasing either parameter shifted the cutoff frequency towards higher frequencies. In all cases, the overall filter characteristics, including passband behaviour and attenuation profile, remained largely unchanged because the component ratios within the filter topology were preserved.

Design Insight

The study demonstrates that active filter circuits can be conveniently re-tuned for different operating frequencies through simple component selection without requiring modifications to the overall circuit topology. This provides significant flexibility in practical analog circuit design, where identical filter architectures are often adapted to different application requirements by selecting appropriate resistor and capacitor values.

Implementation Notes

During the implementation of the active filter circuits, several practical considerations were encountered before obtaining the expected theoretical responses. These observations provided valuable insight into the importance of proper circuit configuration and simulation setup in analog circuit design.

The selection of an appropriate operational amplifier model was found to be an important aspect of the implementation. Initial simulations using practical operational amplifier models produced responses that differed from theoretical expectations due to improper supply configuration. The simulations were subsequently standardized using the **UniversalOpamp2** model powered with **$\pm 15\text{ V}$** supplies, ensuring linear operation over the complete frequency range considered in this work.

Another important observation was the influence of the operational amplifier configuration on overall filter behaviour. The first-order filters required the operational amplifier to operate as a unity-gain voltage follower to isolate the RC network from loading effects. For the second-order filters, the Sallen-Key topology required correct implementation of both the feedback network and power supply connections to achieve the expected second-order frequency response.

Throughout the design process, AC sweep analysis was used as the primary verification tool to validate cutoff frequency, attenuation characteristics and phase response. Transient simulations were subsequently performed to confirm the time-domain behaviour predicted by the frequency-domain analysis.

These implementation considerations demonstrate that successful analog circuit simulation depends not only on theoretical design equations but also on appropriate device selection, circuit topology and simulation configuration.

Conclusion

This project presented the design, simulation and comparative evaluation of first-order and second-order active low-pass and high-pass filters using LTspice. The circuits were designed to investigate the influence of filter topology and order on analog signal-conditioning performance while maintaining a common cutoff frequency of approximately **1 kHz**.

The simulated frequency responses closely matched theoretical predictions, validating the design methodology based on the RC time constant. The first-order filters demonstrated gradual attenuation characteristics suitable for basic frequency-selective applications, whereas the second-order Sallen-Key filters exhibited significantly improved frequency selectivity through steeper stopband

attenuation and increased phase variation. These results clearly illustrate the advantages of higher-order filter implementations for applications requiring enhanced suppression of unwanted frequency components.

Component variation studies further confirmed that the cutoff frequency can be effectively controlled through appropriate selection of resistor and capacitor values without altering the overall filter topology. The implementation also emphasized the importance of selecting a suitable operational amplifier model and providing proper power supply connections to achieve reliable simulation results.

Overall, the project successfully demonstrated the design, analysis and verification of active analog filters using LTspice while providing practical insight into simulation-driven circuit development, frequency response analysis and analog signal-conditioning techniques.

References

Books

1. Sedra, A. S., & Smith, K. C., *Microelectronic Circuits*, Oxford University Press.
2. Boylestad, R. L., *Electronic Devices and Circuit Theory*, Pearson Education.
3. Franco, S., *Design with Operational Amplifiers and Analog Integrated Circuits*, McGraw-Hill Education.

Software

4. Analog Devices, *LTspice XVII Circuit Simulation Software*.

Datasheets / Technical References

5. Analog Devices, *Operational Amplifier Fundamentals and Active Filter Design Notes*.

Appendix

Circuit Parameters Used Throughout the Project:

Parameter	Value
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Operational Amplifier	UniversalOpamp2
Supply Voltage	$\pm 15\text{ V}$
R1	15.9 k Ω
R2	15.9 k Ω
C1	10 nF
C2	10 nF
Cutoff Frequency	$\approx 1\text{ kHz}$